

Open the evidence document, **evidence.doc**

Make sure that your name, centre number and candidate number appear on every page of this document. This document must contain your answers to each question.

Save this evidence document in your work area as:

evidence_ followed by your centre number_candidate number, for example: **evidence_zz999_9999**

A class declaration can be used to declare a record.

If the programming language used does not support arrays, a list can be used instead.

One source file is used to answer **Question 2**. The file is called **QueueData.txt**

- 1 This iterative pseudocode algorithm for the function `IterativeVowels()` takes a string as a parameter and counts the number of lower-case vowels in this string.
The vowels are the letters a, e, i, o and u.

```

FUNCTION IterativeVowels(Value : STRING) RETURNS INTEGER
    DECLARE Total : INTEGER
    DECLARE LengthString : INTEGER
    DECLARE FirstCharacter : CHAR
    Total ← 0
    LengthString ← LENGTH(Value)
    FOR X ← 0 TO LengthString - 1
        FirstCharacter ← MID(Value, 0, 1)
        IF FirstCharacter = 'a' OR FirstCharacter = 'e' OR
           FirstCharacter = 'i' OR FirstCharacter = 'o' OR
           FirstCharacter = 'u' THEN
            Total ← Total + 1
        ENDIF
        Value ← MID(Value, 1, LENGTH(Value)-1)
    NEXT X
    RETURN Total
ENDFUNCTION

```

The pseudocode function `MID(X, Y, Z)` returns Z number of characters from string X, starting at the character in position Y. The first character in a string is in position 0, for example:

`MID("computer", 0, 3)` returns "com"

The pseudocode function `LENGTH(X)` returns the number of characters in the string X, for example:

`LENGTH("computer")` returns 8

- (a) (i) Write program code for the function `IterativeVowels()`.

Save your program as **Question1_N23**.

Copy and paste the program code into part **1(a)(i)** in the evidence document.

[5]

- (ii) Write program code to call the function `IterativeVowels()` with the parameter `"house"` from the main program.

Output the return value.

Save your program.

Copy and paste the program code into part **1(a)(ii)** in the evidence document.

[2]

- (iii) Test your program.

Take a screenshot of the output.

Save your program.

Copy and paste the screenshot into part **1(a)(iii)** in the evidence document.

[1]

- (b) (i) Rewrite the function `IterativeVowels()` as a recursive function with the identifier `RecursiveVowels()`.

Save your program.

Copy and paste the program code into part **1(b)(i)** in the evidence document.

[6]

- (ii) Write program code to call the function `RecursiveVowels()` with the parameter `"imagine"` from the main program.

Output the return value.

Save your program.

Copy and paste the program code into part **1(b)(ii)** in the evidence document.

[1]

- (iii) Test your program.

Take a screenshot of the output.

Save your program.

Copy and paste the screenshot into part **1(b)(iii)** in the evidence document.

[1]